SAP Database Development Report

Final Term Project

SAP Account: BD1-026

ISOM-230: Big Data, Business Intelligence and Analytics

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Suffolk University Spring 2026

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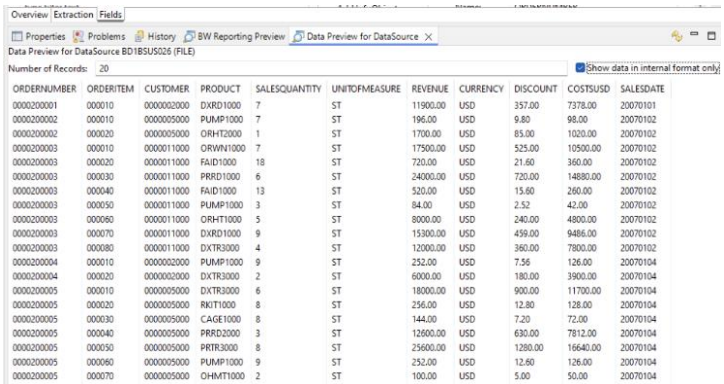
April 27, 2026

Benchmark 2: Data Acquisition

In more common terms, what is a DataSource and how does an InfoPackage interact with a DataSource?

A DataSource is the bridge between a source system and the SAP BW data warehouse. It defines where the data comes from, what it looks like, and how BW should read it. In our project, the DE and US DataSources each pointed to a specific Global Bike sales CSV flat file, making it the Extract step of the ETL process. Additionally, an InfoPackage is the object that triggers and controls the data load into SAP BW. It works by reading the DataSource's metadata to know what to pull and how.



ORDERNUMBER	ORDERITEM	CUSTOMER	PRODUCT	SALESQUANTITY	UNITOFMEASURE	REVENUE	CURRENCY	DISCOUNT	COSTSUSD	SALESDATE
0000200001	000010	0000002000	DXRD1000	7	ST	11900.00	USD	357.00	7378.00	20070101
0000200002	000010	0000005000	PUMP1000	7	ST	196.00	USD	9.80	98.00	20070102
0000200002	000020	0000005000	ORHT2000	1	ST	1700.00	USD	85.00	1020.00	20070102
0000200003	000010	0000011000	ORWN1000	7	ST	17500.00	USD	525.00	10500.00	20070102
0000200003	000020	0000011000	FAID1000	18	ST	720.00	USD	21.60	360.00	20070102
0000200003	000030	0000011000	PRRD1000	6	ST	24000.00	USD	720.00	14880.00	20070102
0000200003	000040	0000011000	FAID1000	13	ST	520.00	USD	15.60	260.00	20070102
0000200003	000050	0000011000	PUMP1000	3	ST	84.00	USD	2.52	42.00	20070102
0000200003	000060	0000011000	ORHT1000	5	ST	8000.00	USD	240.00	4800.00	20070102
0000200003	000070	0000011000	DXRD1000	9	ST	15300.00	USD	459.00	9486.00	20070102
0000200003	000080	0000011000	DXTR3000	4	ST	12000.00	USD	360.00	7800.00	20070102
0000200004	000010	0000002000	PUMP1000	9	ST	252.00	USD	7.56	126.00	20070104
0000200004	000020	0000002000	DXTR3000	2	ST	6000.00	USD	180.00	3900.00	20070104
0000200005	000010	0000005000	DXTR3000	6	ST	18000.00	USD	900.00	11700.00	20070104
0000200005	000020	0000005000	RKIT1000	8	ST	256.00	USD	12.80	128.00	20070104
0000200005	000030	0000005000	CAGE1000	8	ST	144.00	USD	7.20	72.00	20070104
0000200005	000040	0000005000	PRRD2000	3	ST	12600.00	USD	630.00	7812.00	20070104
0000200005	000050	0000005000	PRTR3000	8	ST	25600.00	USD	1280.00	16640.00	20070104
0000200005	000060	0000005000	PUMP1000	9	ST	252.00	USD	12.60	126.00	20070104
0000200005	000070	0000005000	OHMT1000	2	ST	100.00	USD	5.00	50.00	20070104

Figure 1: Data Preview for US DataSource BD1BSUS026

Benchmark 3: Modeling Basics

In more common terms, what is an InfoObject?

An InfoObject is the basic building block in SAP BW and is similar to a column in a database, as it explains how data is stored and used. There are two types: Key Figures, which are numeric values, and Characteristics, which are descriptive categories. In our project, we built both types in Eclipse, and these definitions became the foundation for everything built in later benchmarks, from transformations to the final reports

What items or things do you set to define a key figure or characteristic?

In SAP BW, Key figures and characteristics are defined by characteristics such as data type and numeric/calculation rules. For example, when building the Material InfoObject we enabled Master Data and Texts, set it as Language Dependent, and added four attributes, including Product Category and Division as navigational characteristics. These settings directly ensure data quality consistency across both the US and German datasets used in our BW model.

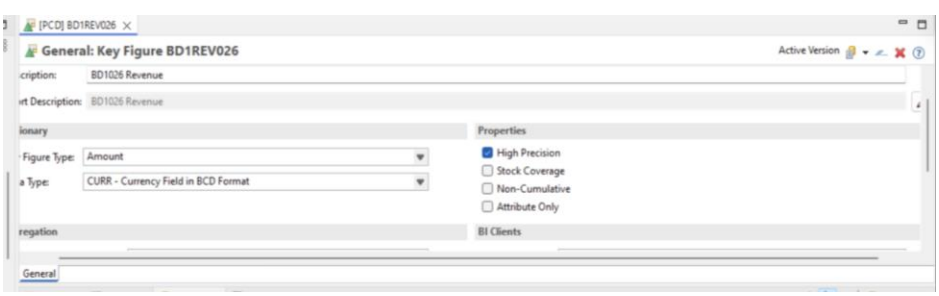


Figure 2: Product Category Characteristic InfoObject BD1PC026 in Eclipse

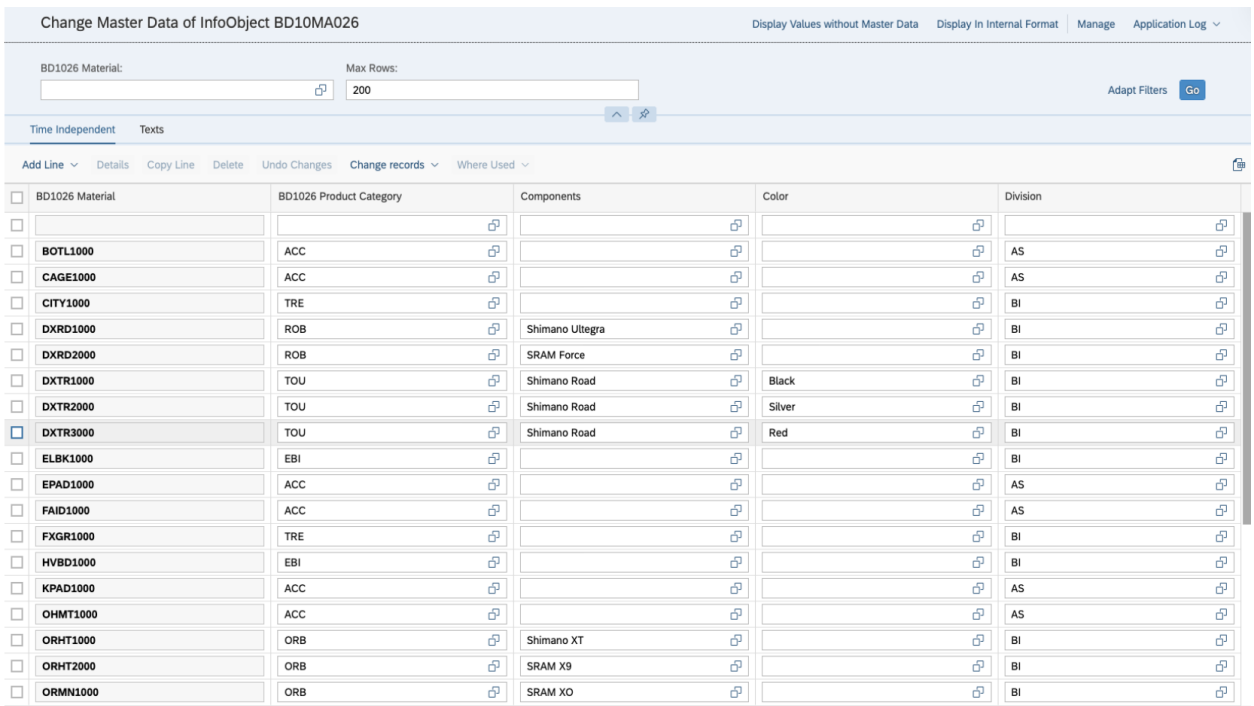
Benchmark 4: Master Data

What is happening in Benchmark 4?

Benchmark 4 is where all three ETL stages were applied together for the first time. We created a Data Flow for Material as the visual blueprint, configured a DataSource pointing to Product.csv, built Transformations mapping source fields to target InfoObject fields, and executed two separate Data Transfer Processes -- one for Material Attributes and one for Texts, since SAP BW stores these in separate master data tables and requires them to be loaded independently.

What is Master Data, how does it differ from Transactional Data, and are different loading steps required?

Master Data describes stable core business entities like Material, Customer, and Product Category that provide context for all warehouse activity and must be loaded first since transaction records depend on them for validation and enrichment. Transactional Data is dynamic, generated continuously from business events, capturing measurable facts like Revenue and Sales Quantity at a specific point in time. Their loading steps reflect this difference: Master Data requires separate Transformations and DTPs per data type (Attributes and Texts) within a single InfoObject, while Transactional Data requires separate DTPs per source system, in our case one for DE and one for US due to their different number and currency formats.



BD1026 Material	BD1026 Product Category	Components	Color	Division
BOTL1000	ACC			AS
CAGE1000	ACC			AS
CITY1000	TRE			BI
DXRD1000	ROB	Shimano Ultegra		BI
DXRD2000	ROB	SRAM Force		BI
DXTR1000	TOU	Shimano Road	Black	BI
DXTR2000	TOU	Shimano Road	Silver	BI
DXTR3000	TOU	Shimano Road	Red	BI
ELBK1000	EBI			BI
EPAD1000	ACC			AS
FAID1000	ACC			AS
FXGR1000	TRE			BI
HVBD1000	EBI			BI
KPAD1000	ACC			AS
OHMT1000	ACC			AS
ORHT1000	ORB	Shimano XT		BI
ORHT2000	ORB	SRAM X9		BI
ORMN1000	ORB	SRAM XO		BI

Figure 4: Verified Material master data attributes successfully loaded into SAP BW/4HANA

Benchmark 5: Data Warehouse Build

How does loading the transactional (TX) data differ from loading Master data and what transformation actions have been taken?

Master Data loading is driven by the internal structure of the InfoObject, requiring separate steps for Attributes and Texts, whereas Transactional Data loading is driven by the number of data source systems. For example, we executed two separate DTPs because our DE and US data came from DataSources with different numbers and currency formats. Additionally, in terms of transformation actions, we used transformations such as Direct Assignment to map fields like Customer straight across with no conversion, and a one-to-many mapping to derive four time characteristics from our single SALESDATE field. We also used a formula-based transformation to calculate Net Sales by subtracting Discount from Revenue rather than reading it directly from the source.

What are the different types of Data Stores and how are InfoObjects used in the definition of Advanced Data Stores

In terms of the different types of Data Stores, SAP BW includes four main types of DSOs: Standard DSO, Write-Optimized DSO, Direct Update DSO, and the aDSO, which we used in this benchmark. In order to define the structure of our aDSO, we used InfoObjects. For example, in this benchmark, we replaced raw DataSource fields with InfoObjects to give each field a standardized business meaning. Once an aDSO is defined by InfoObjects, it serves as the central storage layer where transformed and enriched data are held and made available for reporting and analysis.

What is a DSO and what is a data cube?

A DSO (DataStore Object) stores data at the detailed record level in a flat two-dimensional table, which represented our Data Mart layer, storing our fully transformed and enriched sales transaction records from both Global Bike Germany and US. Additionally, a data cube stores data in a multidimensional star schema structure, optimized for analytical reporting across dimensions like product, region, and time. In this benchmark, our data cube has fewer rows than our aDSO because the data cube aggregates records, whereas our aDSO holds every individual transaction.

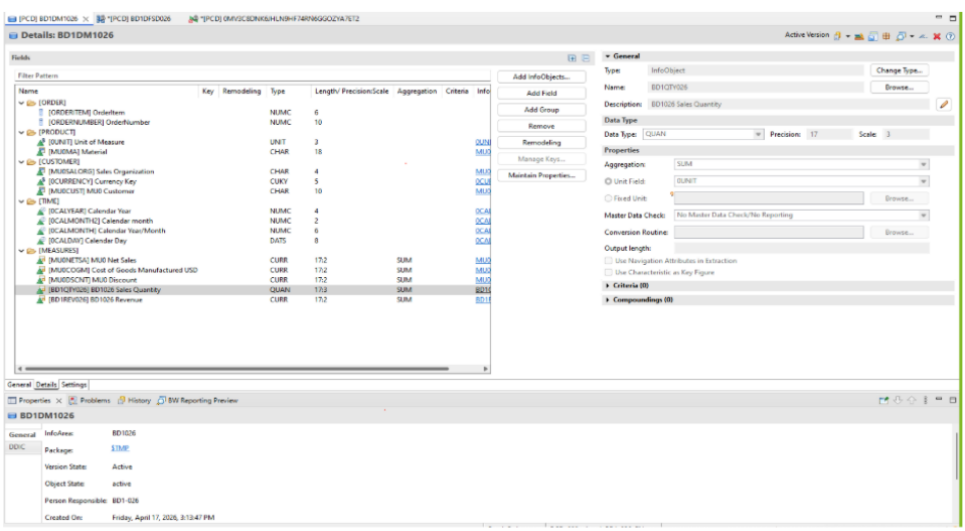
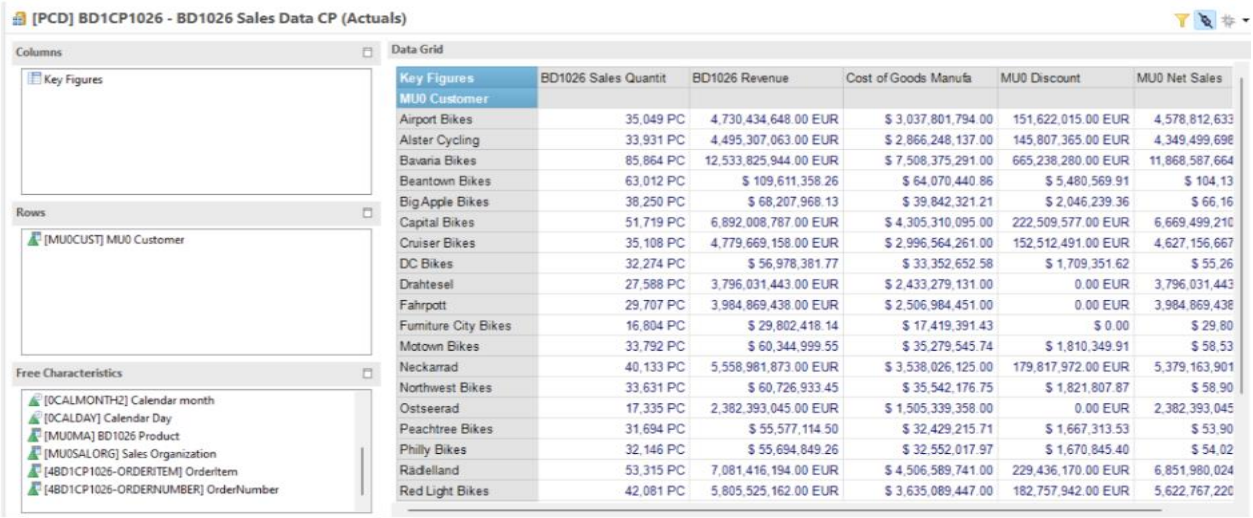


Figure 5: Sales Data aDSO InfoObject Structure (BD1DM1026)

Benchmark 6: Data Warehouse Reporting

In Benchmark 6 we built Composite Provider BD1CP1026 in SAP BW, connecting the Sales Data aDSO to the query layer and exposing all key figures: Revenue, Sales Quantity, Discount, Cost of Goods Manufactured, and Net Sales for analysis in SAP Analysis for Excel.

Do your report totals match the weekly assignment totals? If not, why not?
Our totals do not match. The key figures in our report are inflated across all customers because the DTP was executed more than once during troubleshooting in Benchmark 5. Each execution loads an additional set of records as a new request rather than overwriting existing data, causing duplicate transactions to accumulate and all key figures to be counted multiple times. In production this would be resolved by deleting the duplicate requests from the Data Management screen in Fiori and reloading cleanly with a single DTP execution per source.

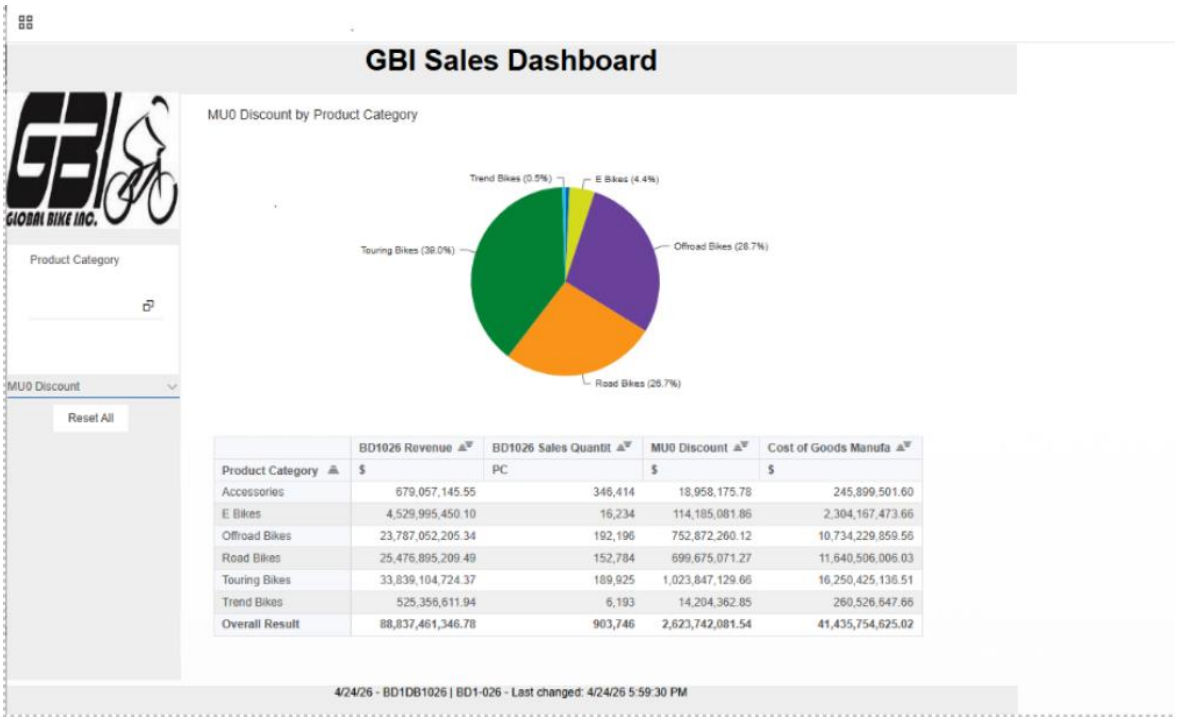


Key Figures / MU0 Customer	BD1026 Sales Quantit	BD1026 Revenue	Cost of Goods Manufa	MU0 Discount	MU0 Net Sales
Airport Bikes	35,049 PC	4,730,434,648.00 EUR	$ 3,037,801,794.00	151,622,015.00 EUR	4,578,812,633
Alster Cycling	33,931 PC	4,495,307,063.00 EUR	$ 2,866,248,137.00	145,807,365.00 EUR	4,349,499,698
Bavaria Bikes	85,864 PC	12,533,825,944.00 EUR	$ 7,508,375,291.00	665,238,280.00 EUR	11,868,587,664
Beantown Bikes	63,012 PC	$ 109,611,358.26	$ 64,070,440.86	$ 5,480,569.91	$ 104,13
Big Apple Bikes	38,250 PC	$ 68,207,968.13	$ 39,842,321.21	$ 2,046,239.36	$ 66,16
Capital Bikes	51,719 PC	6,892,008,787.00 EUR	$ 4,305,310,095.00	222,509,577.00 EUR	6,669,499,210
Cruiser Bikes	35,108 PC	4,779,669,158.00 EUR	$ 2,996,564,261.00	152,512,491.00 EUR	4,627,156,667
DC Bikes	32,274 PC	$ 56,978,381.77	$ 33,352,652.58	$ 1,709,351.62	$ 55,26
Drahtesel	27,588 PC	3,796,031,443.00 EUR	$ 2,433,279,131.00	0.00 EUR	3,796,031,443
Fahrpott	29,707 PC	3,984,869,438.00 EUR	$ 2,506,984,451.00	0.00 EUR	3,984,869,438
Furniture City Bikes	16,804 PC	$ 29,802,418.14	$ 17,419,391.43	$ 0.00	$ 29,80
Motown Bikes	33,792 PC	$ 60,344,999.55	$ 35,279,545.74	$ 1,810,349.91	$ 58,53
Neckarrad	40,133 PC	5,558,981,873.00 EUR	$ 3,538,026,125.00	179,817,972.00 EUR	5,379,163,901
Northwest Bikes	33,631 PC	$ 60,726,933.45	$ 35,542,176.75	$ 1,821,807.87	$ 58,90
Ostseerad	17,335 PC	2,382,393,045.00 EUR	$ 1,505,339,358.00	0.00 EUR	2,382,393,045
Peachtree Bikes	31,694 PC	$ 55,577,114.50	$ 32,429,215.71	$ 1,667,313.53	$ 53,90
Philly Bikes	32,146 PC	$ 55,694,849.26	$ 32,552,017.97	$ 1,670,845.40	$ 54,02
Rädelland	53,315 PC	7,081,416,194.00 EUR	$ 4,506,589,741.00	229,436,170.00 EUR	6,851,980,024
Red Light Bikes	42,081 PC	5,805,525,162.00 EUR	$ 3,635,089,447.00	182,757,942.00 EUR	5,622,767,220

Figure 6: Composite Provider BD1CP1026 Report Output by Customer

Benchmark 7: Dashboards

What is a dashboard, what advantages does it have over a Benchmark 6 report, and what ways can data be conveyed in a dashboard?
In Benchmark 7, we built the GBI Sales Dashboard using SAP Lumira Designer, connecting it directly to our Composite Provider query as a data source. A dashboard is a visual and interactive reporting application that goes beyond a static report like the one in Benchmark 6 by allowing users to explore data dynamically through multiple component types. In our dashboard, data was conveyed through a Crosstab showing Revenue, Sales Quantity, Discount, and Cost of Goods Manufactured by Product Category, and a Chart that displayed the same data visually as a column chart by default and switched to a pie chart when Discount was selected from the dropdown. Users could filter by Product Category using a Dimension Filter, isolate individual key figures using a Dropdown Box, and reset all filters with a single Button, making the dashboard a live analytical tool rather than a passive output.



	BD1026 Revenue	BD1026 Sales Quantit	MU0 Discount	Cost of Goods Manufa
Product Category	$	PC	$	$
Accessories	679,057,145.55	346,414	18,958,175.78	245,899,501.60
E Bikes	4,529,995,450.10	16,234	114,185,081.86	2,304,167,473.66
Offroad Bikes	23,787,052,205.34	192,196	752,872,260.12	10,734,229,859.56
Road Bikes	25,476,895,209.49	152,784	699,675,071.27	11,640,506,006.03
Touring Bikes	33,839,104,724.37	189,925	1,023,847,129.66	16,250,425,136.51
Trend Bikes	525,356,611.94	6,193	14,204,362.85	260,526,647.66
Overall Result	**88,837,461,346.78**	**903,746**	**2,623,742,081.54**	**41,435,754,625.02**

Figure 7: GBI Sales Dashboard with Pie Chart and Interactive Filters (SAP Lumira Designer)

Benchmark 8: Predictive Analytics

Why is data mining important, how does SAP PA compare to R or Python, and what types of models can be built in SAP PA?

In Benchmark 8, we used SAP Predictive Analytics Expert Analytics to perform time series analysis on GBI sales revenue data. Data mining is important because it goes beyond descriptive reporting to uncover hidden patterns and forecast future outcomes, which is exactly what GBI needed since sales revenue had been declining in the US since the 2008 financial crisis. Compared to building the same model in R or Python, which would require writing code to import data, load libraries, specify parameters, and generate visualizations, SAP PA accomplished the entire workflow through a visual point-and-click interface where we imported the sales data, applied a USD filter, and connected the Triple Exponential Smoothing algorithm without writing a single line of code. We chose Triple Exponential Smoothing specifically because the GBI sales data exhibits both a seasonal pattern and a declining trend, and unlike Single or Double Exponential Smoothing, which cannot account for both simultaneously, Triple Exponential Smoothing models all three components of level, trend, and seasonality, which is why it fit the historical data closely and produced a reliable twelve-month forecast for 2012.

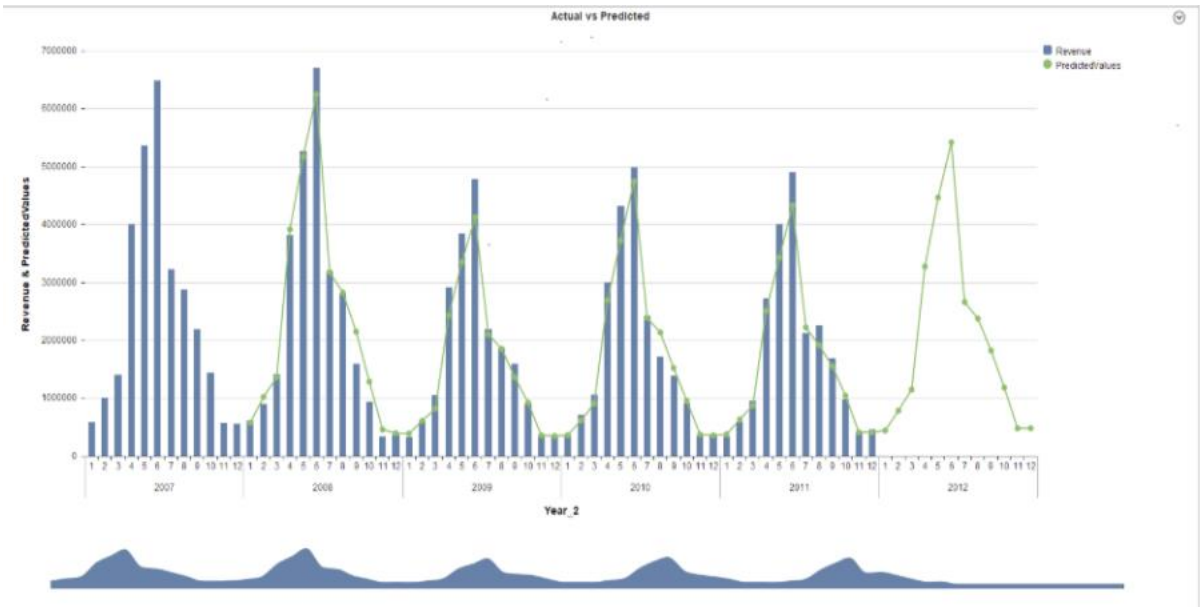


Figure 8: Triple Exponential Smoothing Forecast: Actual vs Predicted GBI Sales Revenue 2007–2012